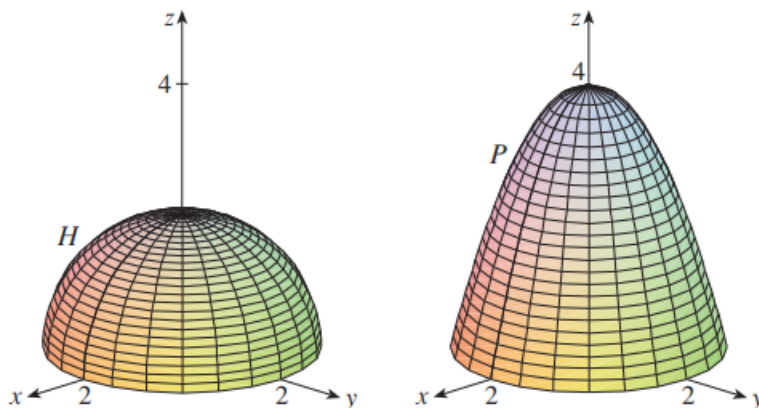


Math 2551 Worksheet: Stokes' Theorem

1. Let H be the hemisphere and P be the portion of a paraboloid shown below. Use Stokes' Theorem to explain why, if $\mathbf{F}$ is a vector field on $\mathbb{R}^3$ whose components have continuous partial derivatives, we must have

$$\iint_H (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, d\sigma = \iint_P (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, d\sigma.$$



2. Use Stokes' Theorem to evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, d\sigma$, where $\mathbf{F} = \langle x^2 z^2, y^2 z^2, xyz \rangle$ and S is the part of the paraboloid $z = x^2 + y^2$ that lies inside the cylinder $x^2 + y^2 = 4$, oriented upward.
3. A particle moves along line segments from the origin to the points $(1, 0, 0)$, $(1, 2, 1)$, $(0, 2, 1)$, and back to the origin under the influence of the force field

$$\mathbf{F}(x, y, z) = z^2 \mathbf{i} + 2xy \mathbf{j} + 2y^2 \mathbf{k}.$$

Find the work done by the field on the particle.